Executive Summary: Versioning & Change Control

This audit report evaluates the MEP Calculator codebase against the FENMO (Globel) Application QA/UAT Checklist for Project Versioning, Author Attribution, Bulk-Edit Engineering & Storage Path Consistency (IDs 5, 35, and 36). It validates snapshot restoration, author metadata, and cell grid bulk operations.

ID	Checklist Item	Severity	Audit Result	Action Required
ID 5	Version Management (Save/Restore/Author Tracking)	High	PASS (100%)	None - VersionManager & author metadata active
ID 35	Edit & Bulk-Edit Change Handling	High	PASS (100%)	None - useBulkActions grid engine operational
ID 36	Version Save Path & Storage Location Consistency	Medium	PASS (100%)	None - Desktop local path & cloud endpoints verified

1. Detailed Component Verification & Evidence

Item ID	Architecture Component	Technical Verification & Evidence
ID 5	Version Manager & Author Tracking File: frontend/src/components/workspace/VersionManager.tsx Endpoints: /api/projects/save-version & /api/projects/restore-version	EVIDENCE: Automatically increments versions (`V0`, `V1`, `V2`...). Injects `_versionMetadata` recording `createdBy`, `createdAt`, `editedBy`, `lastEditedAt`, `user`, and custom engineering description. Restoration safely loads all calculator inputs while preserving authentication and active project context.
ID 35	Grid Engine Bulk Actions Files: frontend/src/core/gridEngine/useBulkActions.ts & useSelection.ts Hooks: bulkDelete, bulkDuplicate, bulkEdit	EVIDENCE: Enables multi-cell/row operations with validation. Updating any field via bulk edit immediately pushes state to history, updates calculation totals, and displays confirmation toasts (`Updated X rows`).
ID 36	Storage Location Consistency Components: Cloud DB + Local Disk (Tauri invoke 'save_backup')	EVIDENCE: Web sessions persist to server database `project_versions.json`. Desktop clients write to the explicit configurable `versionsPath` directory with nested `<ProjectName>/<CalculatorName>/<vX>.json` folders.

2. Version Snapshot Lifecycle & Restoration Workflow

The following diagram illustrates how project revisions are created, tagged with author metadata, and restored without state pollution:

1. Snapshot Creation ('Save As Version')

- User clicks 'Save As Version' in top action bar or Version Manager dropdown.
- Calculates next sequential version code (e.g. V0 -> V1 -> V2).
- Gathers active workspace state (spaces, zones, equipment) and module localStorage items.
- Embeds author signature: `createdBy: currentUserName`, timestamp, and description notes.

2. Multi-Tier Storage Dispatch

- Cloud: Dispatches `POST /api/projects/save-version` to store in Supabase PostgreSQL & server cache.
- Local (Desktop App): Saves standalone JSON to configured backup path `<Project>/<Calculator>/<vX>.json`.

3. Safe Restoration Engine

- User selects historical version from Version History drawer.
- System clears dirty flags and replaces calculator parameters with version snapshot.
- Preserves critical session metadata (`mep_active_project_id`, `auth_token`) to prevent session drops.

3. Audit Conclusion & Compliance Status

All 3 checklist requirements (IDs 5, 35, and 36) under Versioning & Change Control are fully implemented, verified, and 100% compliant with professional design audit standards. No code modifications are required for this module.